

12.6 Discovering Inverse Operations

Lesson Introduction

 Benchmark	MA.6.AR.2.2 Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers.		 Learning Target	I can use manipulatives to discover the role of inverse operations when solving one-step addition and subtraction equations.
 Benchmark Coherence	Building On MA.5.AR.2.4 Given a mathematical or real-world context, write an equation involving any of the four operations to determine the unknown whole number with the unknown in any position.		Working Toward MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.	
 Essential Questions	- What are inverse operations? - How would you use inverse operations to solve an equation?			
 Lesson Narrative	In this lesson, students apply and extend previous understandings of arithmetic to algebraic expressions. Using visual models like algebra tiles and bar models and scales to represent abstract concepts of operations and equations, students explore using addition and subtraction as inverse operations to solve one-step equations and use manipulatives to discover the role of inverse operations to solve one-step equations.			
 Component Summary	12.6.1 Warm-Up (~5 min.)	Create a visual representation to solve a real-world problem (<i>SE p. 307</i>)	12.6.4 Guided Practice (10 min.)	Use inverse operations to solve equations (<i>SE p. 310</i>)
	12.6.2 Exploration (10 min.)	Solve equations using algebra tiles (<i>SE p. 308</i>)	12.6.5 Your Turn! (15 min.)	Solve problems using algebra tiles to model equations (<i>SE p. 311</i>)
	12.6.3 Guided Instruction (5 min.)	Solve equations using inverse operations (<i>SE p. 309</i>)	12.6.6 Wrap-Up (~5 min.)	Write a short note to summarize the lesson (<i>SE p. 312</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Inverse operation <u>Additional Terms:</u> - Opposite - Addition Property of Equality - Subtraction Property of Equality		(optional) Algebra tiles (optional) Blank tape diagrams (optional) Balance scale and marbles	
			Additional Preparation	
			N/A	

 Teacher Tips	Common Misconceptions • Even though “opposites” and “inverse” can be used interchangeably in this lesson, be cautious that students will associate “inverse operations” with “opposites” in all cases. • Students may not have much experience with variables and be confused by a letter standing in for a number. • Students may struggle with integer operations.	Things to Consider • Consistently switch the order of “addition” and “subtraction” in discussion to instill that both operations are important and relate to one another. • Tell students that while “opposites” and “inverse” can be used interchangeably in this lesson, that is not always the case. In mathematics, “opposites” refer specifically to two numbers whose sum is zero. • Solidify mathematical understanding of the terms “opposite” and “inverse” by stating the property being used is the additive inverse or multiplicative inverse. Emphasize that concept of equality by performing the same operation on both sides of the equation. Using arithmetic sentences such as $9 + 6 = 15$ will still be true if 3 is added to both sides resulting in $9 + 6 + 3 = 15 + 3$. If the additive inverse of 6 is added to both sides as in $9 + 6 + -6 = 15 + -6$ the result is still that both sides are equivalent, $9 = 9$, because $6 + -6 = 0$. This should help students remember that the balance between both sides must remain for the equal sign to be used, ensuring students understand the meaning of the equal sign.	
	Literacy and Language Routines Notice and Wonder 12.6.2 Exploration positions students to use algebra tiles to represent and solve algebraic equations. Consider applying Notice and Wonder routines on each set of tiles, positioning students to discover the value of the variable through observation of the algebra tiles.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.2.1: Multiple Representations Students will be working to solve one-step equations algebraically. Clarifications within this standard emphasize building understanding through modeling and using manipulatives. Utilize the progression of the lesson to highlight the process by which students make connections between concept and representation.	
	 Differentiate Instruction	12.6.2 Exploration uses algebra tiles to show inverse operations. Consider allowing students to use manipulative algebra tiles as a path to hands-on learning. Encourage students to explain their understandings and the processes they use to reach them.	
Lesson Notes:			

12.6.1 Warm-Up: Picture This

Component Overview: Students draw a visual representation of a mathematical situation and then use the image to solve the problem. Encourage students to explain the reasoning they used to solve the problem.

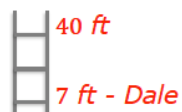


12.6.1 Warm-Up: Picture This

Dale is helping to fix the roof at his uncle's house. They are using a 40-foot ladder. Dale is climbing the ladder to reach the roof. He is currently 7 feet up the ladder.

1. Sketch a picture to represent this situation.

Answers will vary.



2. How much farther does Dale have to go to reach the roof? Include units.

33 feet



After students complete the Warm-Up, have them explain how they arrived at their answer. This entry point might engage auditory learners. Use sentence stems to encourage conversation such as:

- "I drew..."
- "I figured out..."

12.6.2 Exploration: Solving Equations Using Algebra Tiles

Component Overview: Students use algebra tiles to understand the process of using inverse operations to solve algebraic equations. Students compare the number of algebra tiles on each side of an equation to determine the value of the variable tile. Students then discuss the equivalent relationship with a partner.



12.6.2 Exploration: Solving Equations Using Algebra Tiles

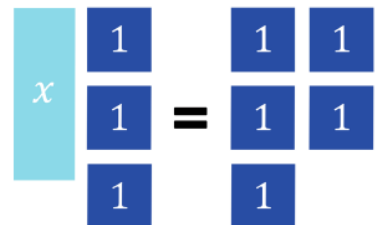
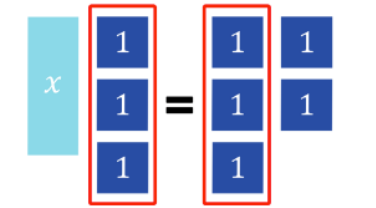
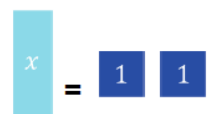
Algebra tiles can be used to represent and solve algebraic equations.

In the model shown,  represents 1 and  represents x .

- The steps to determine the value of x using algebra tiles are shown.
 - Work with your partner to complete the last step according to the description.



Consider allowing students to use algebra tiles to recreate the problem. Prompt students to discuss their rationale and approach as they work to continue building mathematical literacy.

Algebra Tiles	Description
	Use algebra tiles to represent the equation $x + 3 = 5$.
	Determine any tiles that are represented on both sides of the equation.
	Remove the equal parts from both sides of the equation.

12.6.2 Exploration: Solving Equations Using Algebra Tiles (continued)

- b. The expressions $x + 3$ and 5 are equivalent according to the given equation. Discuss with your partner whether the tiles you drew on each side of the equation mat in the last step represent an equivalent relationship. Then, write a brief summary of your discussion.

Answers will vary. The tiles I drew represent an equivalent relationship because the original problem $x + 3 = 5$ represented an equivalent relationship, and since I removed the same number of tiles from each side of the equation, the relationship will still be equivalent.

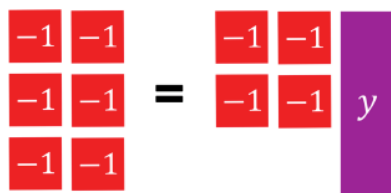
- c. How many blue tiles are equivalent to one teal tile?

2 blue tiles are equivalent to one teal tile.

- d. What is the value of x in the equation $x + 3 = 5$?

$x = 2$

2. In the model shown,  represents -1 and  represents y .



- a. Work with your partner to determine how many red tiles are equivalent to one purple tile.

2 red tiles are equivalent to one purple tile.

- b. What is the value of y in the equation $-6 = -4 + y$?

$-2 = y$



Consider giving students a blank tape diagram. Have students use the diagram to build a model of the equation. Encourage students to explain how and why they are using the pictured algebra tiles to create a bar diagram.



Prompt students to make connections between their visual models and the learning targets for this lesson.

- *What do you notice about the equation?*
- *How does the equation relate to the tiles?*

12.6.3 Guided Instruction: Solving Equations Using Inverse Operations

Component Overview: Students are introduced to the concept of using inverse operations to solve one-step equations. Students begin exploring the inverse operations of addition and subtraction.



12.6.3 Guided Instruction: Solving Equations Using Inverse Operations

An equation relates two equivalent expressions. When solving an equation, the mathematical sentence must remain balanced so that the two expressions remain equivalent.

1. Discuss with your partner what it means for an equation to remain balanced. Then, summarize your discussion.

Answers will vary. For an equation to remain balanced, you must add or subtract exactly the same quantity from both the left and the right side of the equation.

When finding the solution to an equation, inverse operations are used to balance an equation.

An **inverse operation** is an operation that reverses the effect of another operation.

2. Using the definition of inverse operations, complete the table to indicate which operation would reverse the given operation.

Operation	Inverse Operation
Subtraction (–)	<i>Addition (+)</i>
Addition (+)	<i>Subtraction (–)</i>



Work with students to define inverse operations. Discuss how the word “inverse” can be used in other contexts. Review the relationship of addition to subtraction to help students understand why inverse operations work when solving an equation.

Consider showing students inverse operations using arithmetic operations. Emphasize the role of the equal sign.

12.6.4 Guided Practice: Using Inverse Operations to Solve Equations

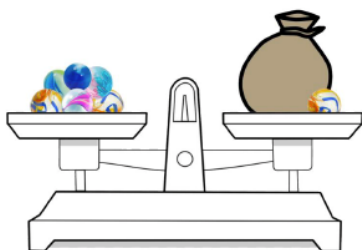
Component Overview: Students use the image of a balance to extend their understanding of inverse operations. Students will also use algebra tiles to model a one-step equation and then explain the process they used to solve it.



12.6.4 Guided Practice: Using Inverse Operations to Solve Equations

Balance scales are used to measure weight, or mass. When both pans are at the same height, that means the objects in the two pans weigh the same amount. In mathematics, balance scales are often used to show balanced equations where the expressions on either side of the balance are equivalent.

1. Heidi and Pete were given the balance scale shown below which has exactly 6 marbles in the left pan. Heidi and Pete were asked how many marbles are in one bag.



Heidi stated, "There is not enough information to solve this problem."

Pete stated, "Each bag holds 5 marbles."

- a. Explain who is correct.

Pete is correct. Answers will vary. The pans are at the same height; therefore, the objects in one pan weigh the same amount as the objects in the other pan. Since there are 6 marbles in one pan of the balance, and 1 bag and 1 marble in the other pan, the bag must contain 5 marbles in order for both pans to weigh the same amount.



Notice and Wonder

Engage the class in a "Notice and Wonder" routine before releasing students to do the guided practice. Ask students what they notice about the balance scale and what they wonder about the balance scale. For students who do not have experience with balances, relate the balance scale to a see-saw.



Consider having students create one-step equations using a balance scale and marbles. Students can place an unequal number of marbles in each pan and then determine how many marbles need to be added or taken away to balance the equation. Help students make the connection between the equations modeled by the marbles and balance and then solving a one-step equation.

12.6.4 Guided Practice: Using Inverse Operations to Solve Equations (continued)

- b. Write an algebraic equation to represent the balance scale problem Heidi and Pete were given.

$$6 = x + 1$$

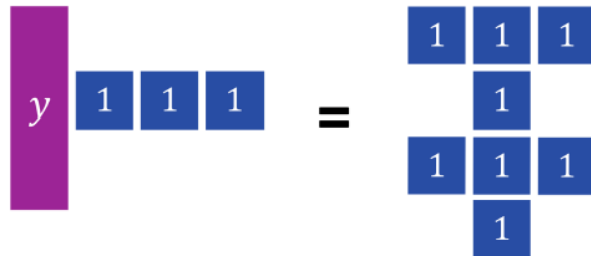
- c. How many marbles are in one bag?

There are 5 marbles in one bag.

- d. What inverse operation can be used to solve the equation?

Subtraction

2. The algebra tiles shown are used to model an equation.



- a. How many blue tiles are equivalent to one purple tile?

5 blue tiles are equivalent to one purple tile.

- b. Write a one-variable equation to represent the tiles on the mat.

$$y + 3 = 8$$

- c. What inverse operation can be used to solve the equation?

Subtraction

- d. What value of y makes the equation true?

$$y = 5$$



For students who are struggling, further scaffolding questions could be:

- What do you notice about the number of blue tiles that are equivalent to one purple tile and the value of y that makes the equation true?
- How can you use inverse operations to solve an equation?

Other supports could include asking student to “cover up” the 1 tiles on the side with the variable y tile and then to “cover-up” the same amount on the right side. What operation would allow those tiles to be removed?

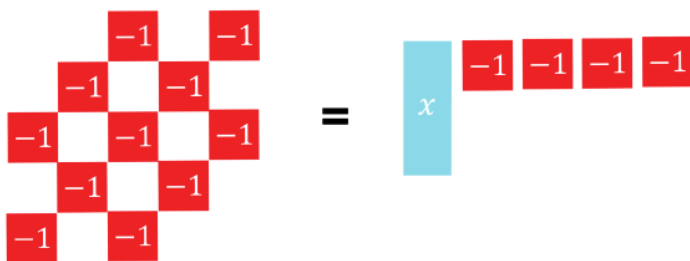
12.6.5 Your Turn!

Component Overview: Students practice using algebra tiles and inverse operations to solve one-step equations.



12.6.5 Your Turn!

1. The algebra tiles shown are used to model an equation.



- a. How many red tiles are equivalent to one teal tile?

7 red tiles are equivalent to one teal tile.

- b. Write a one-variable equation to represent the tiles on the mat.

$$-11 = x - 4$$

- c. What inverse operation can be used to solve the equation?

Addition

- d. What value of x makes the equation true?

$$-7 = x$$



As students work on 12.6.5 Your Turn! make notes on where students are struggling. For some students understanding of the concept will come easily and they will become impatient with the representations. Other students will struggle with the idea of balance and using properties of equality to determine the value of the variable.

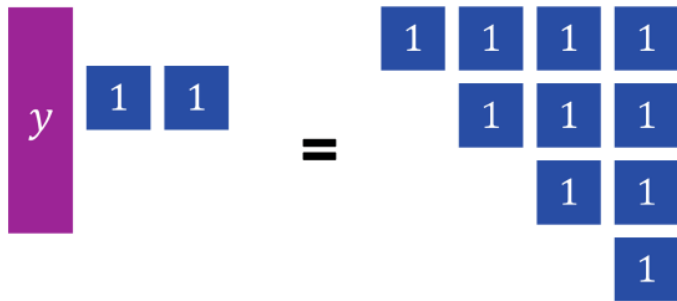
Monitor student answers for question 1b. Some students will use addition of a -4 while other students will use subtraction. The student's choice will likely mean the answers to question 1c are different. Listen for student confusion and be prepared to support the class by pulling them back together as a whole group or addressing these issues during Lesson 7.



Consider having students recreate the equation using manipulative algebra tiles. Students may also benefit from working with a partner and explaining how they solve the problem by moving the algebra tiles in addition to using inverse operations.

12.6.5 Your Turn! (continued)

2. The algebra tiles below model an equation.



Consider an inquiry-based learning approach to facilitate connections between visual models and abstract concepts.

- How can you figure out the relationship of blue tiles to purple tiles?
- How can you use this relationship to solve the equation?
- Why do you use inverse operations to solve the equation?

- a. How many blue tiles are equivalent to one purple tile?

8 blue tiles are equivalent to one purple tile.

- b. Write a one-variable equation to represent the tiles on the mat.

$y + 2 = 10$

- c. What inverse operation can be used to solve the equation?

Subtraction

- d. What value of y makes the equation true?

$y = 8$

12.6.6 Wrap-Up: Cool Down

Component Overview: Students will write a description of today's lesson. Students' notes should define inverse operations and capture the idea of using inverse operations to solve a one-step equation.



12.6.6 Wrap-Up: Cool Down

1. Write a short note to a friend who missed class summarizing today's lesson.

Answers will vary. Today's lesson was about using algebra tiles and inverse operations to solve one-step equations.



For students who might struggle representing their conceptual understanding through writing, consider alternative ways to assess or measure learning for this lesson. Students could choose to record themselves explaining the answer verbally, drawing and labeling, or providing an example. All forms can show an equal level of rigor and conceptual understanding.